

Javier Alejandro Ojeda Vargas

Postdoctoral Researcher • Visiting Professor • External Consultant

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RESEARCH STATEMENT

Ph.D. in Earth Sciences specialized in Geophysics/Seismology Applied to Geological Hazard Assessment and Engineering Geology. My research integrates geotechnical characterization, strong-ground-motion simulation, and seismic hazard assessment, with direct applications in civil and mining works (tailings dams, slopes, critical infrastructure). I also developed a research line in active tectonics and aseismic deformation, applying seismological and satellite geodesy tools. This work has allowed me to collaborate with groups from Chile, France, Italy, Japan, the United States, and Spain, consolidate international networks, diversify my research lines, and secure project funding, external financing, travel grants, and conference invitations. My productivity is summarized in 11 WoS articles, 6 articles under review, 30+ conference presentations, 10+ years of university teaching, student supervision, consulting for companies linked to mining projects, and participation in scientific education programs.

EDUCATION

α =THESIS PROJECT IN CO-TUTELLE

Ph.D. in Earth and Environmental Sciences, IPGP, Université Paris Cité [α]	Oct 2021 - Dec 2024
Ph.D. in Geology, Departamento de Geología, Universidad de Chile [α]	Mar 2020 - Dec 2024
M.Sc. in Geophysics, Departamento de Geofísica, Universidad de Chile	Mar 2017 - Sep 2018
B.S. in Geophysics, Departamento de Geofísica, Universidad de Chile	Mar 2011 - Jun 2016

RESEARCH WORK EXPERIENCE

Postdoctoral Researcher, Anillo EASER project, Universidad de Concepción 	Since January 2025
Visiting Professor, Departamento de Ingeniería Civil, Universidad de Concepción 	Since August 2025
Researcher, Programa de Riesgo Sísmico, Universidad de Chile 	Jul 2024 - Dec 2024
Part-time Research Assistant, Departamento de Geofísica, Universidad de Chile 	Oct 2020 & Jun 2022
Part-time Research Assistant, Programa de Riesgo Sísmico, Universidad de Chile 	Apr 2016 - Aug 2019
Technical assistant, Fondecyt programs 1170430, 11130230 	2016 & 2018
Intern, Centro Sismológico Nacional, Universidad de Chile 	Apr 2015

RESEARCH VISITS

Barcelona Supercomputing Center, Collaborator: Natalia Zamora	May 2026 - Jun 2026
University of California Berkeley, Collaborator: Roland Bürgmann	Sep 2024 - Oct 2024
University of California Berkeley, Supervisor: Roland Bürgmann	Dec 2023 - Mar 2024
Université Côte d'Azur, Supervisor: Jean-Mathieu Nocquet	Nov 2022 - Dec 2022
Istituto Nazionale di Geofisica e Vulcanologia, Supervisor: Aybige Akinci & Elisa Tinti	Jul 2019 & Dec 2019
Institute de Physique du Globe de Paris, Supervisor: Fabian Bonilla	Nov 2017 - Feb 2018

AWARDS, GRANTS, AND FUNDING

SSA Community Grant, Organization of workshop: 'Training on DAS'	2026
Research grant, SZ4D Science Community Meeting; NSF; AccelNet program 2301732	2026
Funding (Postdoc), EASER: Evolution Assessment of Seismic Risk, ANID	2024-2027
Funding (co-PI), High-frequency strong motion simulations: from small to large earthquakes, INGV	2024-2025
Research grant, SZNet Chile & US Exchange Program; SZ4D; NSF; AccelNet program 2301732	2024
International mobility grant, Service des Études Doctorales de l'IPGP	2023
ANID doctoral scholarship & benefits: operating expenses, cotutelle, internship, extensions	2020 - 2024
Multiple travel grant for conferences, JASSO & SEELA, DGF, DGL, AGU, LACSC	2017 - 2026
Research grant for short stay in France, Universidad de Chile	2017
Outstanding student award, Universidad de Chile	2014 & 2015
Academic excellence scholarship, Universidad de Chile	2011 - 2016
Financial assistance grant, Ministry of Education of Chile	2011 - 2016

PEER-REVIEWED PUBLICATIONS

†=EQUAL AUTHORSHIP CONTRIBUTION

- [11] Kondo, Y., Obayashi, M., Sugioka, H., Shiobara, H., Ito, A., Shinohara, M., Iwamori, H., Kinoshita, M., Miller, M., Tassara, C., **Ojeda, J.** (2024). **Seismic Image of the Central to Southern Andean Subduction Zone Through Finite-Frequency Tomography.** *J. Geophys. Res.: Solid Earth*, 129(11), e2024JB028844.
- [10] Vigny, C., Klein, E., **Ojeda, J.** (2024). **In search for the lost truth about the 1922 & 1918 Atacama earthquakes in Chile.** *J. South Am. Earth Sci.*, 143, 104983.
- [9] Flores, C., **Ojeda, J.**, Otarola, C., Arriola, S., Ruiz, S. (2023). **Stochastic strong ground motion simulation in the Santiago Metropolitan region considering an Mw 7.8 intraplate intermediate-depth earthquake.** *J. South Am. Earth Sci.*, 130, 104501.
- [8] Ito, A., Shiobara, H., Miller, M., Sugioka, H., **Ojeda, J.**, Tassara, C., Shinohara, M., Kinoshita, M., Iwamori, H. (2023). **Long-term Array Observation by Ocean Bottom Seismometers at the Chile Triple Junction.** *J. South Am. Earth Sci.*, 124, 104285.
- [7] **Ojeda, J.**, Morales-Yáñez, C., Ducret, G., Ruiz, S., Grandin, R., Doin, M.-P., Vigny, C., Nocquet, J.-M. (2023). **Seismic and aseismic slip during the 2006 Copiapó swarm in North-Central Chile.** *J. South Am. Earth Sci.*, 123, 104198.
- [6] Tissandier, R., Nocquet, J.-M., Klein, E., Vigny, C., **Ojeda, J.**, Ruiz, S. (2023). **Afterslip of the Mw8.3 2015 Illapel earthquake imaged through a time-dependent inversion of continuous and survey GNSS data.** *J. Geophys. Res.: Solid Earth*, 128(2), e2022JB024778.
- [5] **Ojeda, J.**, Akinci, A., Tinti, E., Arriola, S., Ruiz, S. (2021). **Hybrid Broadband Strong-Motion Simulation to investigate the near-source Characteristics of the M6.5, 30 October 2016 Norcia, Italy Earthquake.** *Soil Dyn. Earthq. Eng.*, 149, 106866.
- [4] **Ojeda, J.**, Ruiz, S. (2021). **Seismic noise variability as an indicator of urban mobility during the COVID-19 pandemic in the Santiago metropolitan region, Chile.** *Solid Earth*, 12, pp. 1075-1085.
- [3] Lecocq, T., et al., including **Ojeda, J.** (2020). **Global quieting of high-frequency seismic noise due to COVID-19 pandemic lockdown measures.** *Science*, 369(6509), 1338-1343.
- [2] **Ojeda, J.**, Ruiz, S., del Campo, F., Carvajal, M. (2020). **The 1960 May 21 Mw 8.1 Concepción earthquake: A deep megathrust foreshock that started the 1960 central-south Chilean seismic sequence.** *Seismol. Res. Lett.*, 91(3), 1617-1627.
- [1] Ruiz, S.[†], **Ojeda, J.[†]**, Otarola, C., Pastén, C., Silva, R. (2018). **Stochastic strong motion simulation in borehole and on surface for the Mw 9.0 Tohoku-Oki 2011 mega-earthquake considering P, SV and SH amplification transfer functions.** *Bull. Seismol. Soc. Am.*, 108(5A), 2333–2346.

Furthermore, articles are currently under review (3), and other projects are submitted or in preparation (14).

SEMINARS AND CONFERENCES (SELECTED)

FULL LIST: 64 PRESENTATIONS, S=SEMINARS, C=CONFERENCES

- [S.2] **Interaction between fast and slow earthquakes along the Chilean subduction zone.** *Active Tectonics Meetings, EPS, University of California Berkeley* (22/10/2024)
- [S.1] **Seismic and aseismic slip during the 2006 Copiapó swarm in North-Central Chile.** *Mardi Séismes, Géoazur, Université Côte d'Azur* (29/11/2022)
- [C.2] **Tracking the Aseismic Slip History Along the Chilean Subduction Zone (18°S-40°S): GPS Observations from 2006 to 2024.** *Slow-to-Fast Earthquake Workshop 2025.*
- [C.1] **Global search of short-term transient in cGPS time series.** *AGU Annual Meeting 2023.* [Invited Talk]

FIELD WORK EXPERIENCE

Seismology , Biobío Chile, ORSCA project FONDEQUIP EQM 160015 🌐	May, Oct 2025
Geodesy , Atacama Chile, Project: S5 project ANR-19-CE31-0003 🌐	Nov 2023
Ocean Floor Seismology , Chile Triple Junction, EPIC & JAMSTEC project 🌐	Feb 2021
Exploration Geophysics , Antofagasta Chile, INSUD project ACT172002, PIA-Anillo, Chile 🌐	Oct 2018
Seismology , Tarapacá Chile, PICTURES project MGL1610, NSF 🌐	Nov 2016

OUTREACH EXPERIENCE

Fundación Geonautas , Co-founder of non-profit organization.	Since July 2021
Rocadictos Radio Show , Member of the editorial committee and panelist	Dec 2021 - Apr 2024
Explora ANID , Scientific advisor, collaborator and coordinator	2016 - 2024
Tatemblando Podcast , Co-creator of podcast on seismology	2020 - 2023
Geociencias para Terrícolas , Member in project of social network dissemination	2019 - 2021
Geociencias para Políticas Públicas , Member in project of vinculation with territorial issues	2016 - 2018

UNIVERSITY TEACHING & MENTORING EXPERIENCE

[DETAILS IN CERTIFICATE](#)

During my recent appointment as a visiting professor, I had the opportunity to teach several courses at the Universidad de Concepción, including teaching in **Natural Risk Assessment** (16 hours), **Applied Seismology** (85 hours), **Geomatics** (4 hours), and **Earthquake-Resistant Design** (8 hrs).

From 2014 to 2024, I served as a teaching assistant in multiple undergraduate courses—covering Seismology, Applied Seismology, Ordinary Differential Equations, and Introduction to Meteorology and Oceanography—and in graduate-level courses such as Theoretical Seismology, Earthquakes in Chile, and GNSS Applications in Earthquake Research. Additionally, I was a teaching assistant for the Postgraduate Diploma in Seismology at the Universidad de Chile for five years. I also designed and taught a new graduate-level course aimed at strengthening students' skills in science communication and the organization of scientific events (Geophysics Postgraduate Workshop). More information is available in the following certificate: [Certificate](#) .

I have also served on the evaluation committee and supervised master's students in Engineering Sciences, Geophysics, and Civil Engineering, working on broad topics such as **Earthquake Engineering**, **Mining Seismology**, and **Volcano Seismology**. Currently, I'm supervising an intern from France who is working on her project in seismology and physics.

SCIENTIFIC SERVICE AND PROFESSIONAL ACTIVITIES

I currently serve as **Director of Seismology** at the Colegio de Geofísicas y Geofísicos de Chile A.G. (since May 2025) and **President of Fundación Geonautas** (since January 2025) . Previously, I was an **Early Career Scientist co-representative for the EGU Seismology Division** from September 2020 to May 2022 , where I contributed to strengthening the global geoscience community, and also I served as an **Executive Committee Member of SZ4Grads Andes Sur** (from June 2023 to May 2025) . I have also served as a convener at national and international conferences, with a focus on bridging the multiscale and temporal gap among Earth Science disciplines.

Beyond these leadership and service roles, I also contribute as an **external reviewer** in scientific, academic, and outreach settings, including peer-review of journal articles (AGU: Earth and Space Science, Tectonophysics, SRL, BSSA, Terra Nova, and Seismica), **evaluation of international funding programs** (SZNet from NSF, USA), and participation in regional research initiatives.

Additionally, I have **coordinated numerous scientific conferences, seminars, and webinars** for various institutions. These include the recent Seismic Hazard Congress in Chile (2025), the SZ4Grads Andes Sur events and a broader talk series (2024), scientific meetings for the AndesNet network (2023), and postgraduate events such as the V Jornada de Postgrado en Geofísica at the Universidad de Chile. Earlier activities include coordinating the Congrès Des Doctorant.es at École Doctorale 560 IPGP in 2022, GeoCharlas T3 in 2020 , and several colloquia and postgraduate events at the Universidad de Chile since 2016.

PROFESSIONAL MEMBERSHIPS

Colegio de Geofísicas y Geofísicos de Chile A.G.	desde 2024
Alumni Université Paris Cité	desde 2025
Alumni Universidad de Chile ID: 92833	desde 2025
Seismological Society of America ID: 510-525-5474	desde 2024
American Geophysical Union ID: 793445 	desde 2018
European Geosciences Union ID: 569323	2021 - 2022